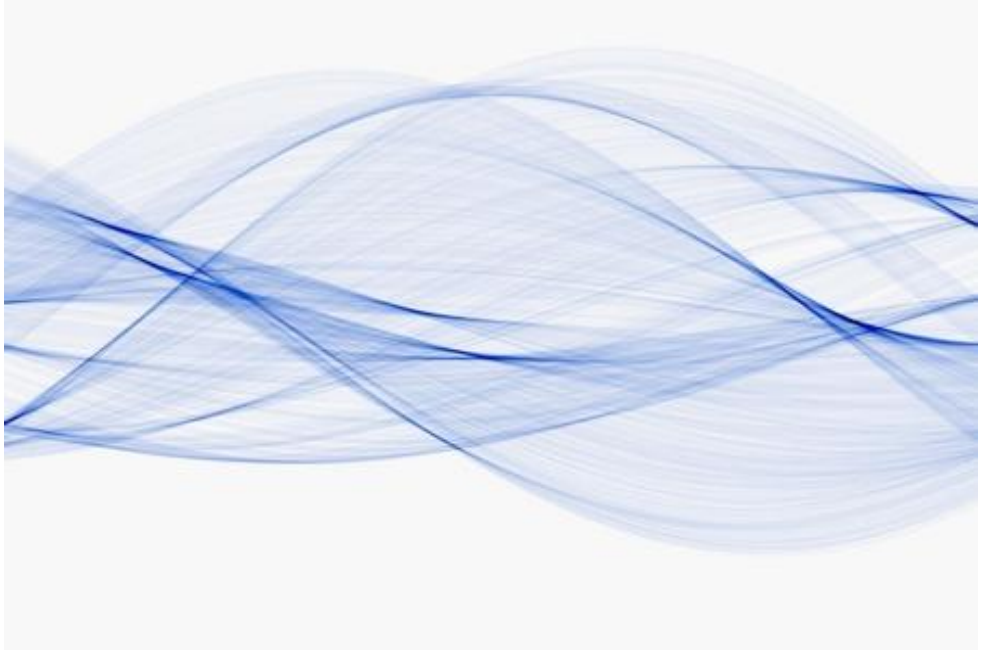
Transmission Characteristics of Optical Fibers

Dr. Monir Morshed
Professor
Dept. of ICT, MBSTU



Transmission Characteristics

- Attenuation,
- Absorption,
- Scattering losses,
- Bending loss,
- Dispersion,
- Material Dispersion
- Waveguide dispersion
- Intra modal dispersion,
- Inter modal dispersion.

Attenuation

- **It is a measure of decay of signal strength or loss of light power that occurs as light pulses propagate through the length of the fiber.**
- **It is mainly caused by absorption and scattering losses.**
- **Absorption is because of fiber material**
- **Scattering is due to structural imperfection within the fiber.**
- **Nearly 90% attenuation happens due to Rayleigh Scattering.**
- **Microbending also contributes to the attenuation of signal.**

Assume a fiber of length L. The input optical power is P_{in}. The optical power is attenuated to P_{out} at the end of the fiber. We define an **attenuation coefficient** α for the fiber.

$$dP = -\alpha P dx$$

$$\frac{dP}{P} = -\alpha dx$$

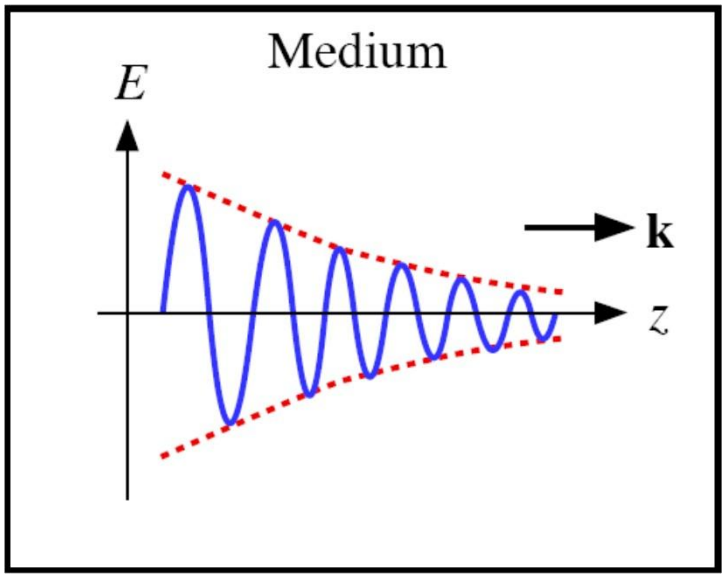
$$\alpha = \frac{1}{L}\ln\left(\frac{P_{in}}{P_{out}}\right)$$

$$P_{out} = P_{in}\exp(-\alpha L)$$

Optical power attenuation in optical fibers is generally expressed in terms of **decibels** per unit length of fiber, typically as dB per km.

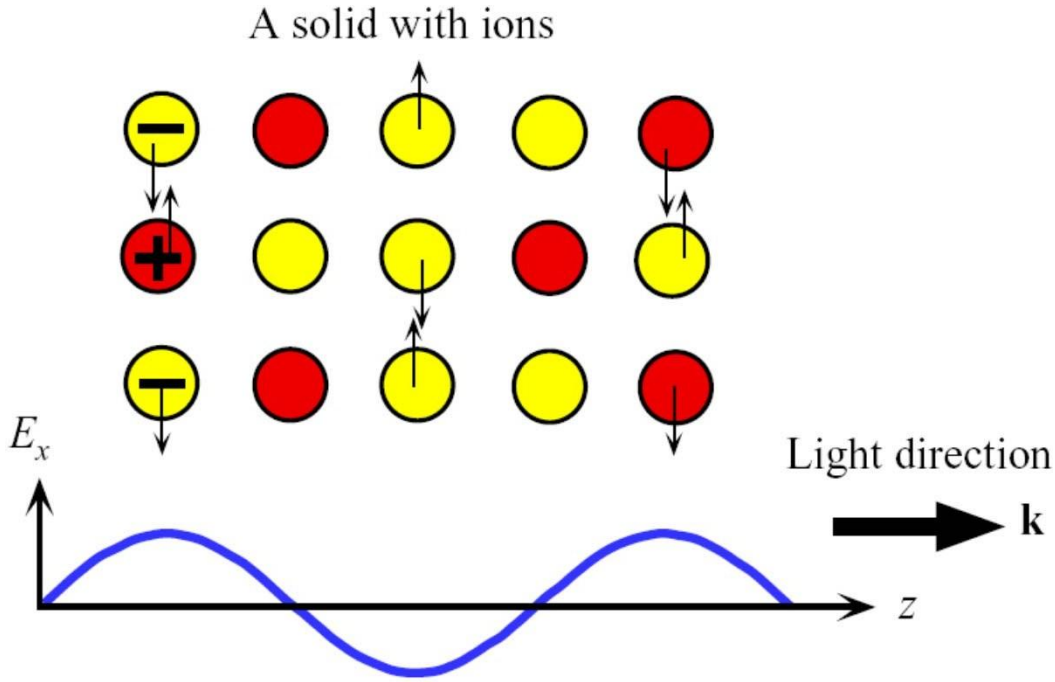
$$\alpha_{dB} = \frac{1}{L}10\log\left(\frac{P_{in}}{P_{out}}\right)$$

Absorption

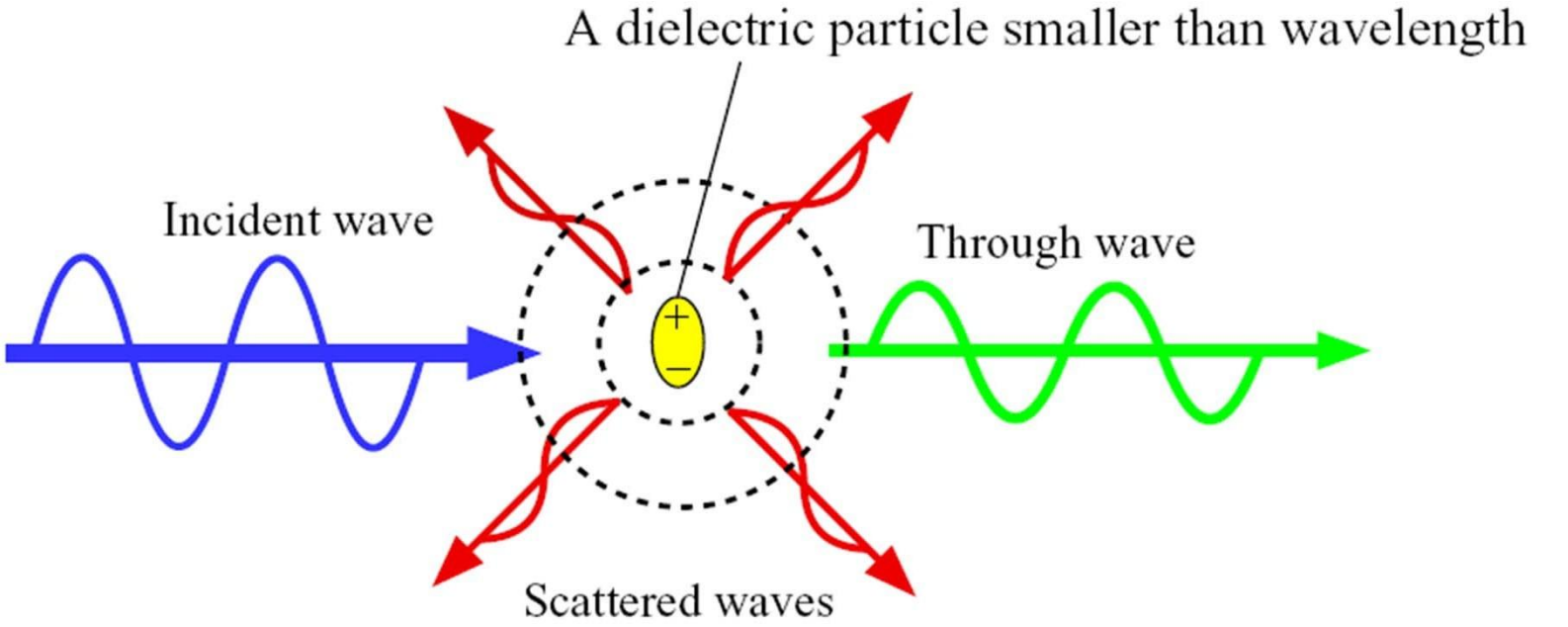


In general, light propagating through a material becomes *attenuated* in the direction of propagation.

In absorption, some of the energy from the propagating wave is converted to other forms of energy, for example, to heat by the generation of lattice vibrations.



Scattering



When a propagating wave encounters a small dielectric particle or a small inhomogeneous region whose refractive index is different from the average refractive index of the medium, the field forces dipole oscillations in the dielectric particle or region by polarizing it, leading to the emission of electromagnetic waves in many directions so that a portion of the light energy is directed away from the incident beam.

Scattering

Whenever the size of a scattering region, whether an inhomogeneity or a small particle, is much smaller ($<\lambda/10$) than the wavelength of the incident wave, the scattering process is generally termed **Rayleigh scattering**.



Lord Rayleigh, an English physicist (1877 – 1919) and a Nobel laureate (1904), made a number of contributions to wave physics of sound and optics.

Attenuation in Optical Fibers

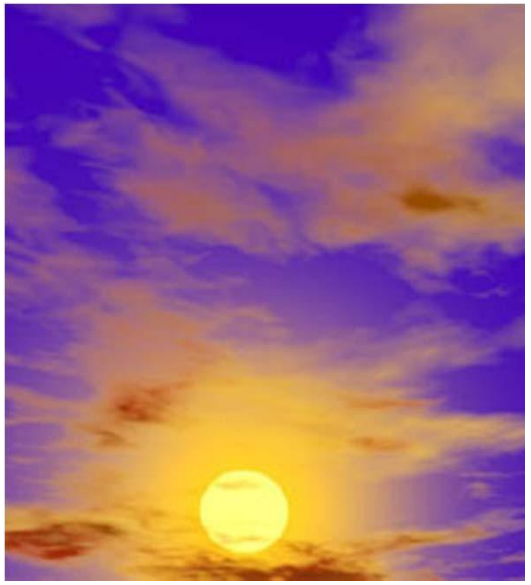
The attenuation α_R in a single component glass due to Rayleigh scattering is approximately given by

$$\alpha_R \approx \frac{8\pi^3}{3\lambda^4}\left(n^2 - 1\right)^2 \beta_T k_B T_f$$

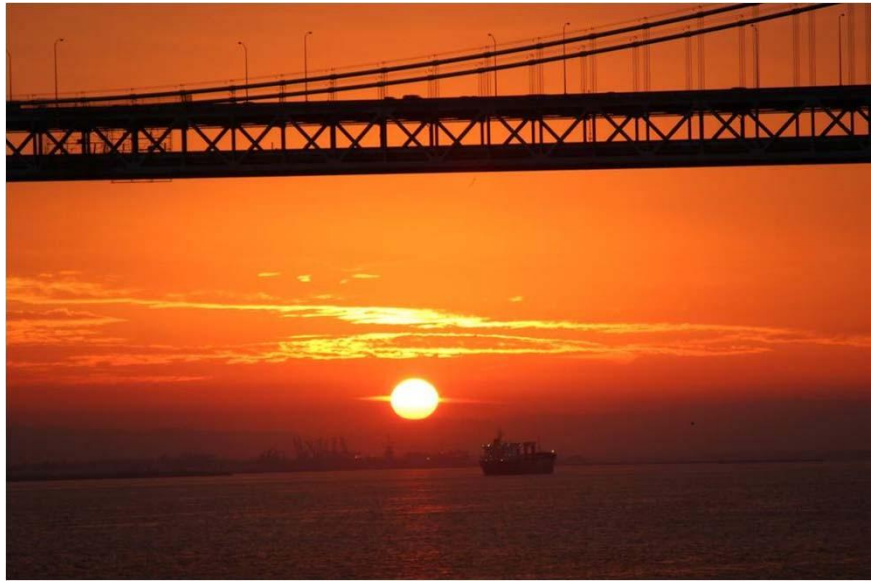
λ is the free space wavelength. T_f is called the **fictive temperature**, at which the liquid structure during the cooling of the fiber is frozen to become the glass structure. β_T is the isothermal compressibility of the glass at T_f.



blue sky



yellow sun

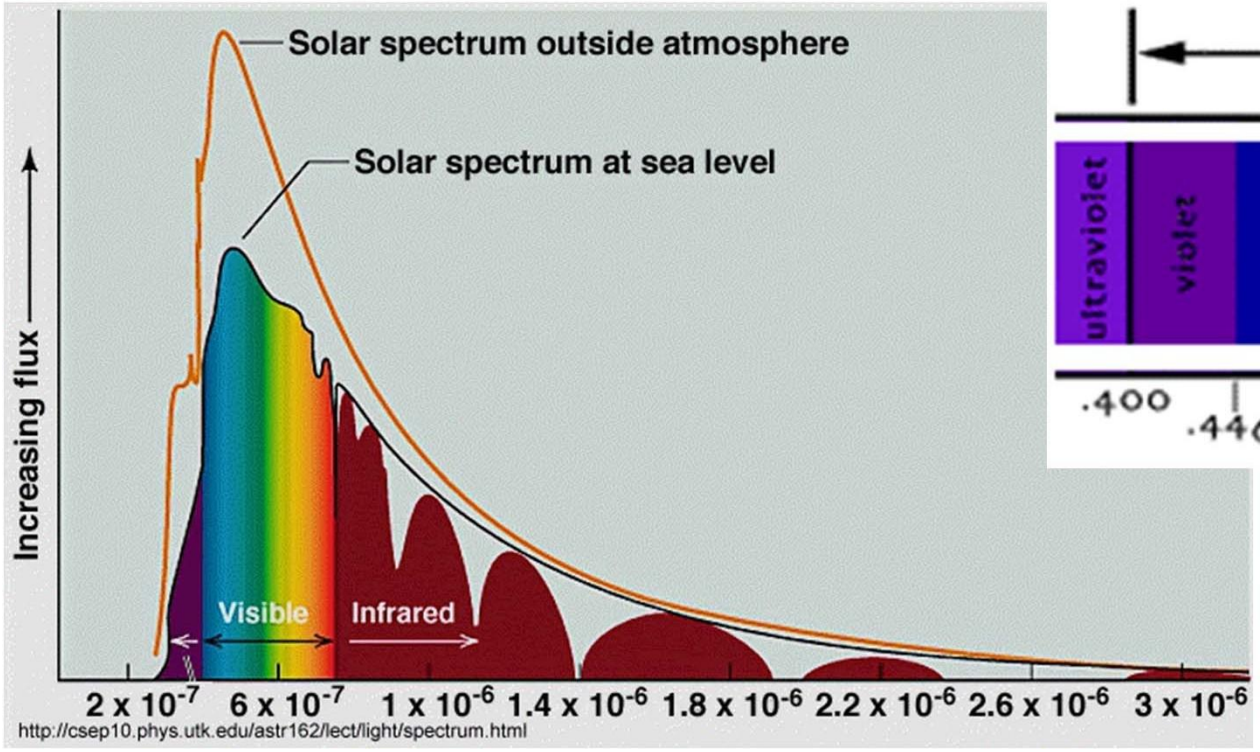


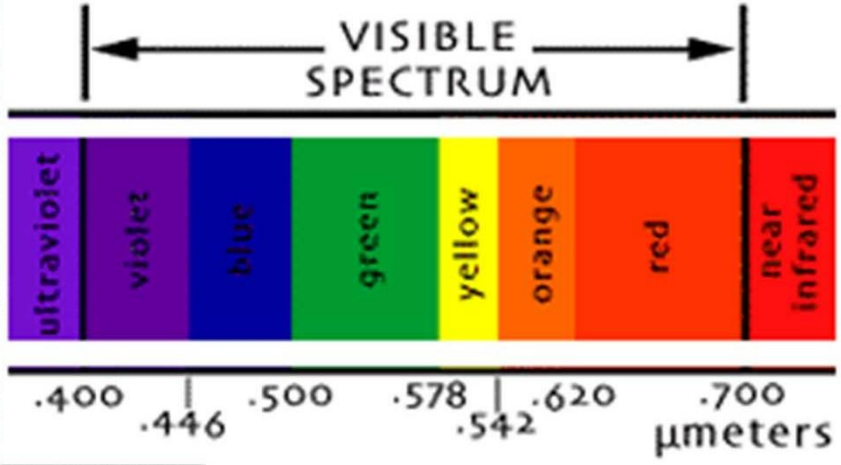
sunrise



sunset

- Why is the sky blue?
- Why does the sun look yellow if we look at the sun directly?
- Why does the sky around the sun appear red during sunrise and sunset?





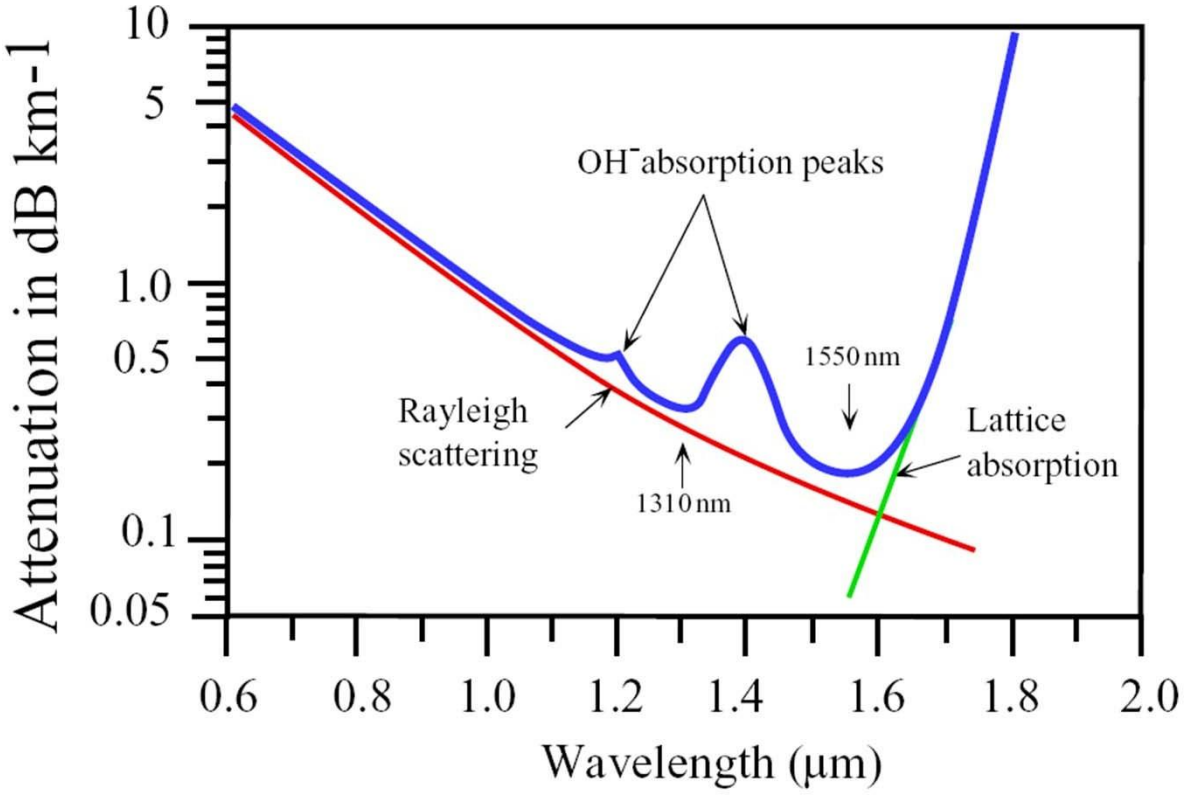
Rayleigh scattering becomes more severe as the frequency of light increases (the wavelength decreases). Blue light that has shorter wavelengths than red light is scattered more strongly by particles in air.

When we look at the sun directly, it appears yellow because the blue light has been scattered.

When we look at the sky in any direction but the sun, our eyes receive scattered light, which appears blue.

At sunrise and sunset, the rays from the sun have to travel the longest distance through the atmosphere and have the most blue light scattered, which gives the sky around the sun its red color at sunrise and sunset.

Attenuation in Optical Fibers



- The sharp increase in attenuation at wavelengths beyond 1.6 μm is due to the energy absorption by the lattice vibrations of silica.
- Two peaks at 1.4 μm and 1.24 μm are due to OH^- ions in silica glass.
- The overall background is due to Rayleigh scattering because of the amorphous structure of silica glass (impossible to eliminate Rayleigh scattering in glass).

Attenuation Problems Example

A low loss fiber has average loss of 3 dB/km at 900 nm. Compute the length over which – a) Power decreases by 50 % b) Power decreases by 75 %.

Solution: $\alpha = 3\ dB/km$

When, power decreases by 50%

$$\alpha_{dB} = \frac{1}{L} 10\log\left(\frac{P_{in}}{P_{out}}\right)$$

$$3 = \frac{1}{L} 10\log(0.5)$$

$$3L = -3$$

$$L = -1$$

Here, negative sign indicates the loss over the length of fiber. So, the length of the fiber is 1 Km.

For a 30 km long fiber attenuation 0.8 dB/km at 1300nm. If a 200 µwatt power is launched into the fiber, find the output power

When mean optical power launched into an 8 km length of fiber is 12 µW, the mean optical power at the fiber output is 3 µW. Determine –

1) Overall signal attenuation in dB.
2) The overall signal attenuation for a 10 km optical link using the same fiber with splices at 1 km intervals, each giving an attenuation of 1 dB.

Attenuation Problems Example

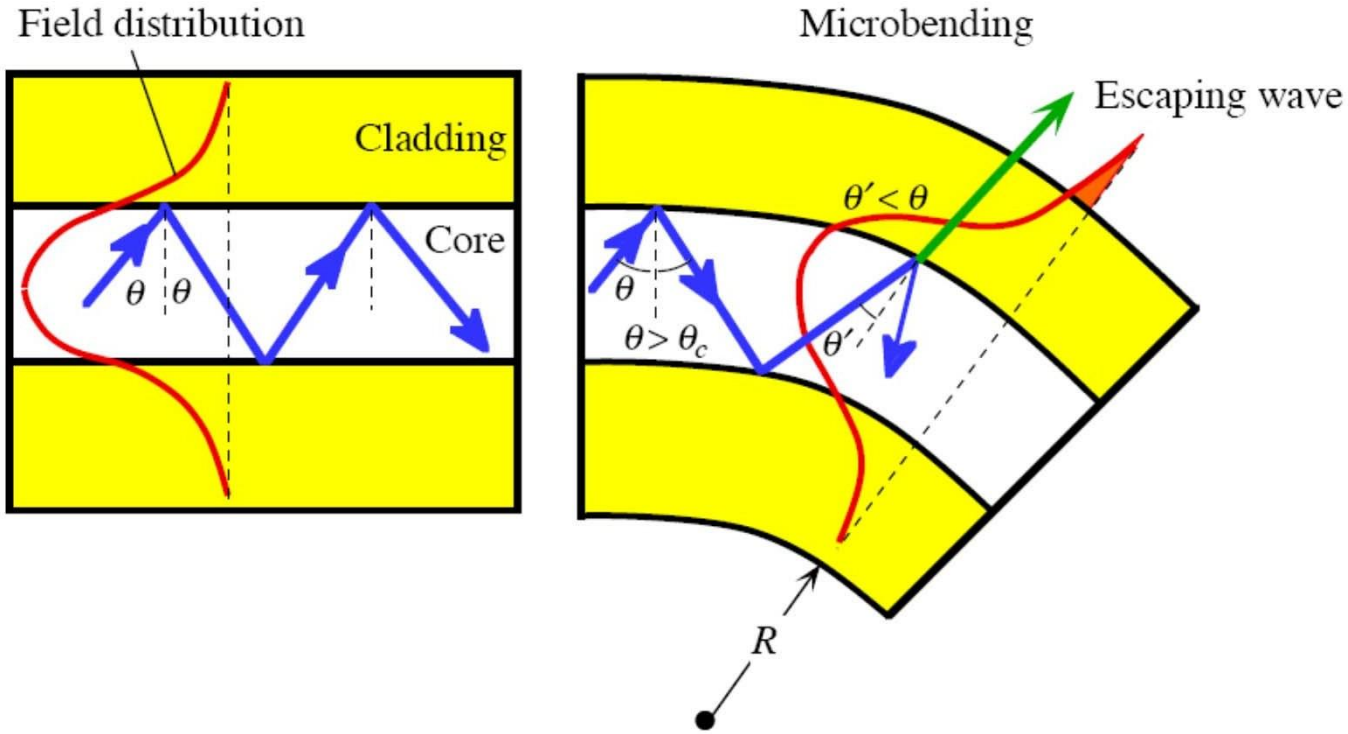
A continuous 12 km long optical fiber link has a loss of 1.5 dB/km. i) What is the minimum optical power level that must be launched into the fiber to maintain as optical power level of 0.3 µW at the receiving end? ii) What is the required input power if the fiber has a loss of 2.5 dB/km?

Optical power launched into fiber at transmitter end is 150 µW. The power at the end of 10 km length of the link working in first windows is – 38.2 dBm. Another system of same length working in second window is 47.5 µW. Same length system working in third window has 50 % launched power. Calculate fiber attenuation for each case and mention wavelength of operation.

The input power to an optical fiber is 2 mW while the power measured at the output end is 2 µW. If the fiber attenuation is 0.5 dB/km, calculate the length of the fiber.

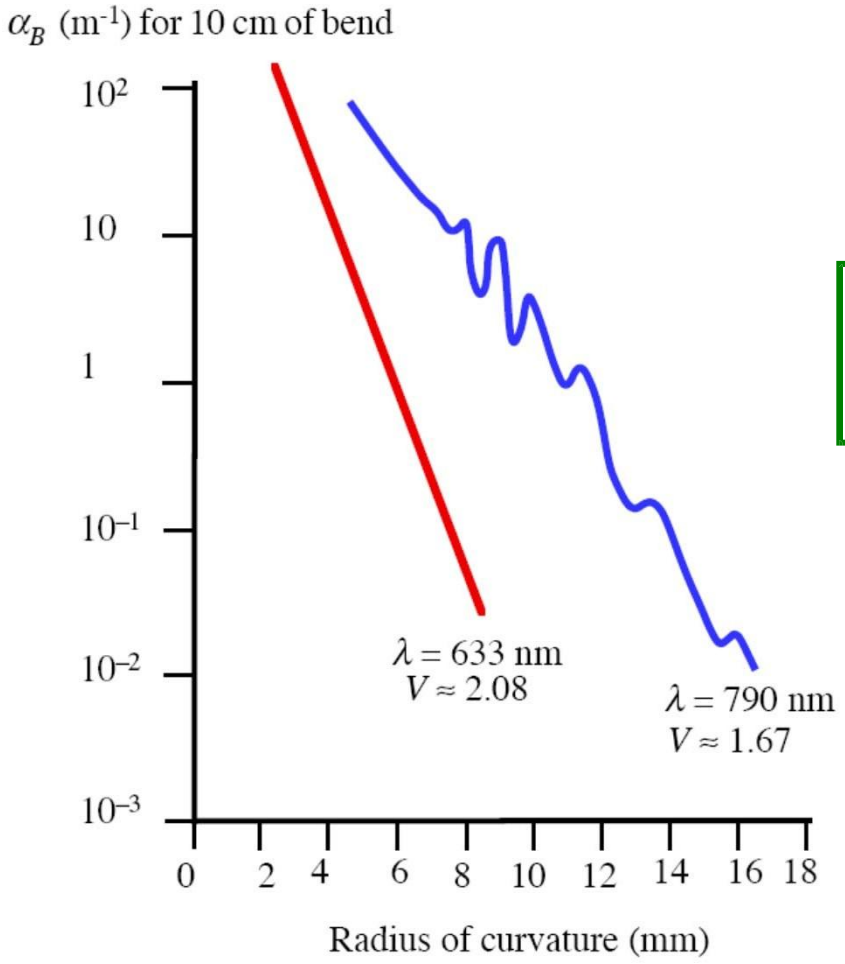
Microbending and Macrobending Losses

Microbending is due to a sharp local bending of the fiber that changes the guide geometry and refractive index profile locally, which leads to some of the light energy radiating away from the guiding direction.



Local bending leads to an decrease in the incidence angle, which induces either an increase in the penetration depth into the cladding or a loss of total internal reflection.

Microbending and Macrobending Losses



Microbending loss increases sharply with decreasing radius of curvature.

Measured microbending loss for a 10-cm fiber bent with different amounts of radius of curvature. Single mode fiber with a core diameter of 3.9 μm, cladding radius 48 μm, $\Delta = 0.00275$, NA ≈ 0.11, $V \approx 1.67$ and 2.08.

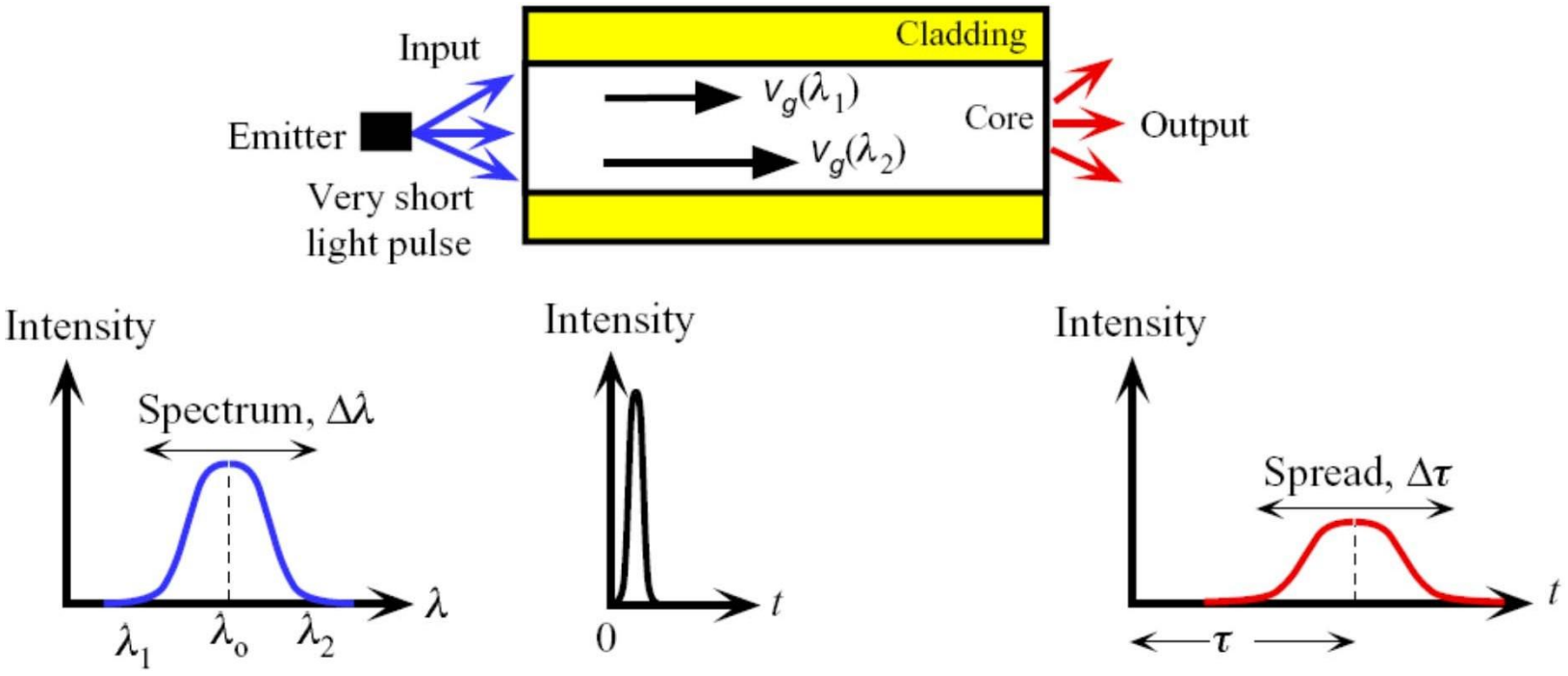
Microbending and Macrobending Losses

Macrobending loss is due to small changes in the refractive index of the fiber due to induced strain when it is bent during its use, for example, when it is cabled and laid. Typically, when the radius of curvature is close to a few centimeters, macrobending loss crosses over into the regime of microbending loss.

Material Dispersion

There is no intermodal dispersion in single-mode fibers.

Material dispersion is due to the dependence of the refractive index on the free space wavelength.



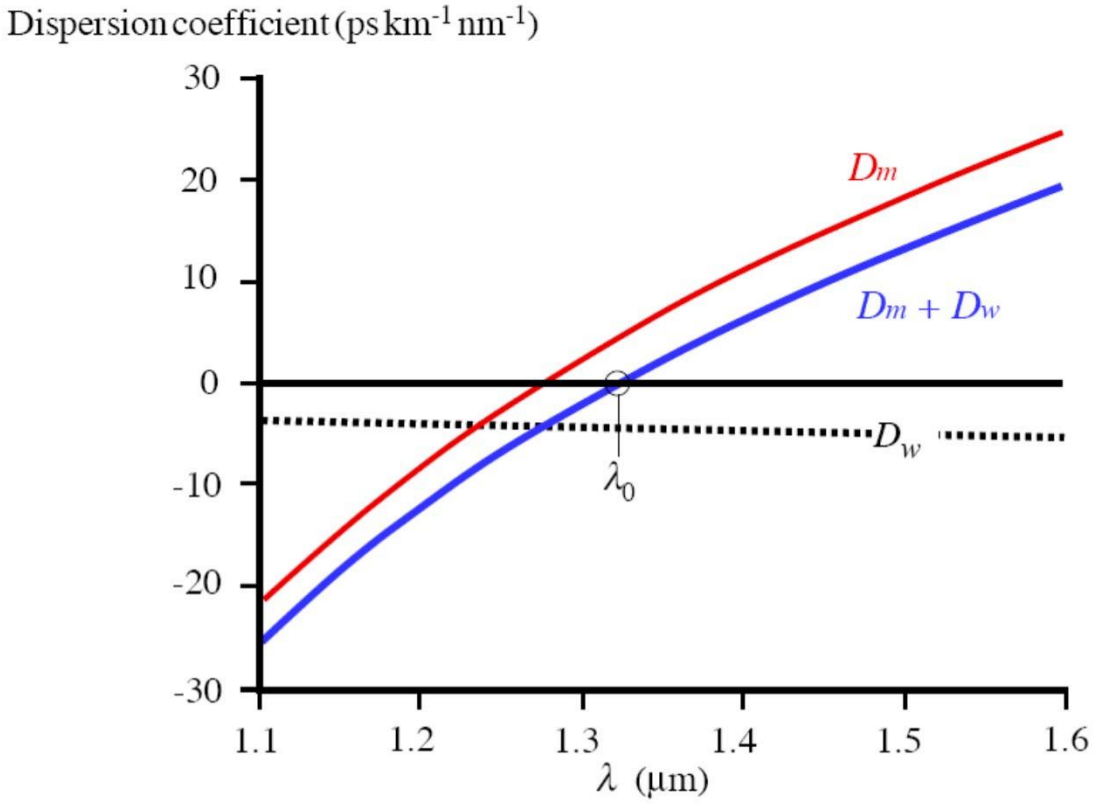
$$\frac{\Delta\tau}{L} = |D_m|\Delta\lambda$$

D_m is called the **material dispersion coefficient**.

Dispersion is expressed as the temporal spread per unit length because slower waves fall further behind the faster waves over a longer distance.

Material Dispersion

$$D_{\mathrm{m}} \approx -\frac{\lambda}{c}\left(\frac{d^2 n}{d\lambda^2}\right)$$



Group Velocity

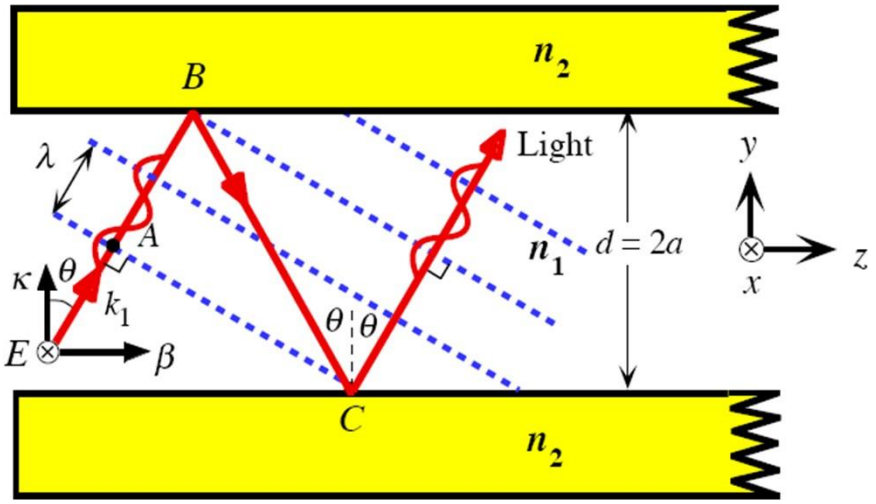
$$\textbf{Group velocity} = v_g = \frac{d\omega}{dk}$$

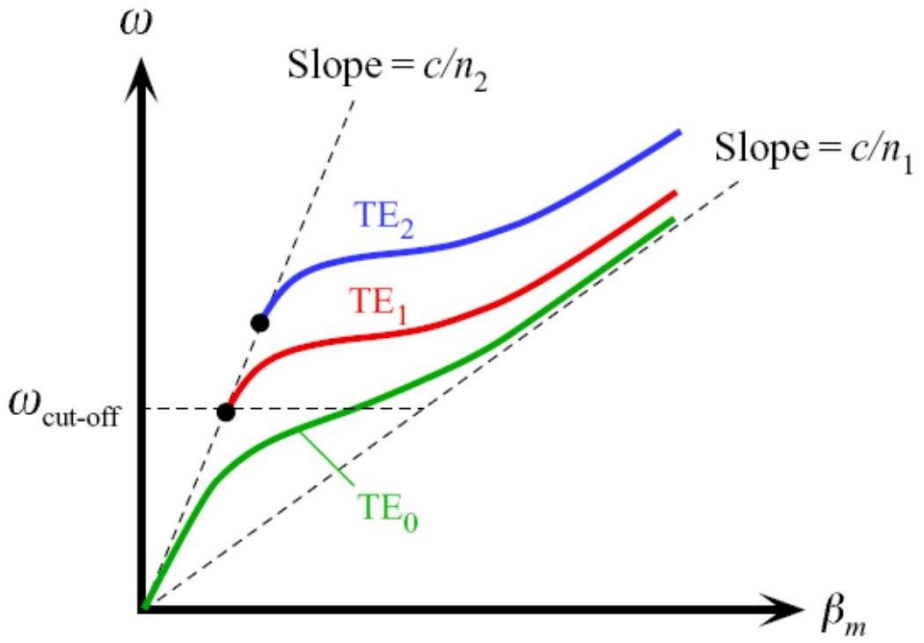
$$\textbf{Phase velocity} = \frac{\omega}{k} = v = \frac{c}{n}$$

The group velocity defines the speed with which energy or information is propagated because it defines the speed of the envelope of the amplitude variation.

To find out the group velocity, we need to know the change of ω with respect to *k*. The ω *versus k* characteristics is called the **dispersion relation** or **dispersion diagram**.

Waveguide Dispersion Diagram





Waveguide condition

$$\left[\frac{2\pi n_1(2a)}{\lambda}\right]\cos\theta_m - \phi_m = m\pi$$

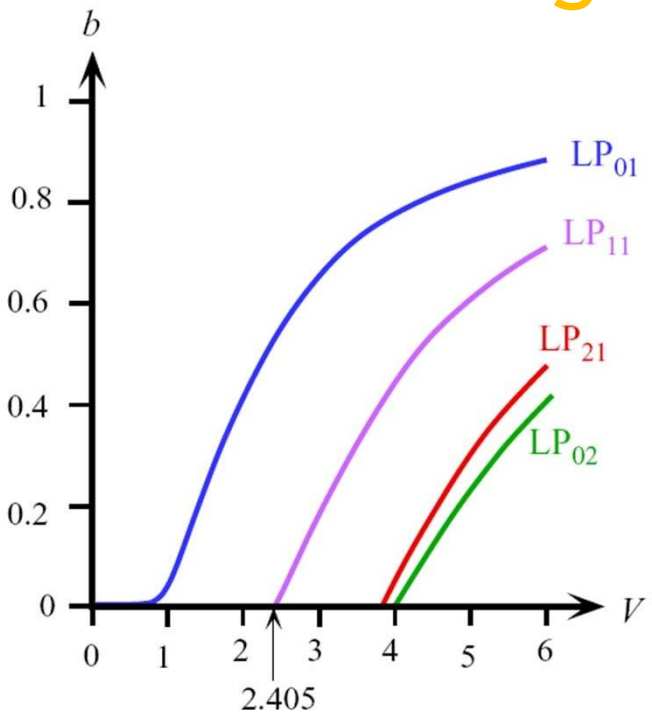
θ_m depends on the waveguide properties (n_1, n_2, and a) and the light frequency, ω.

$$\beta_m = k_1 \sin\theta_m = \frac{2\pi n_1}{\lambda}\sin\theta_m$$

The slope $d\omega/d\beta_m$ at any frequency is the group velocity v_g.

ω is a function of β_m — waveguide dispersion diagram.

Waveguide Dispersion



$$b = \frac{\left(\frac{\lambda\beta}{2\pi}\right)^2 - n_2^2}{n_1^2 - n_2^2}$$

$$V = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2}$$

Waveguide dispersion is due to that the group velocity $d\omega/d\beta$ varies as a function of λ.

$$\frac{\Delta\tau}{L} = |D_w|\Delta\lambda$$

D_w is called the **waveguide dispersion coefficient**.

Profile Dispersion

There is an additional dispersion mechanism called the **profile dispersion** that arises because the group velocity of the fundamental mode, $v_{g,01}$, also depends on the normalized index difference. Δ is dependent on the wavelength due to material dispersion characteristics, *i.e.*, n_1 *versus* λ and n_2 *versus* λ behavior. Therefore, in reality, the profile dispersion originates from the material dispersion.

$$\Delta = \frac{n_1 - n_2}{n_1} \qquad \frac{\Delta\tau}{L} = |D_p|\Delta\lambda$$

D_p is called the **profile dispersion coefficient**.

D_p is less than 1 ps km^{-1} nm^{-1}, much smaller than D_m and D_w.

Chromatic Dispersion

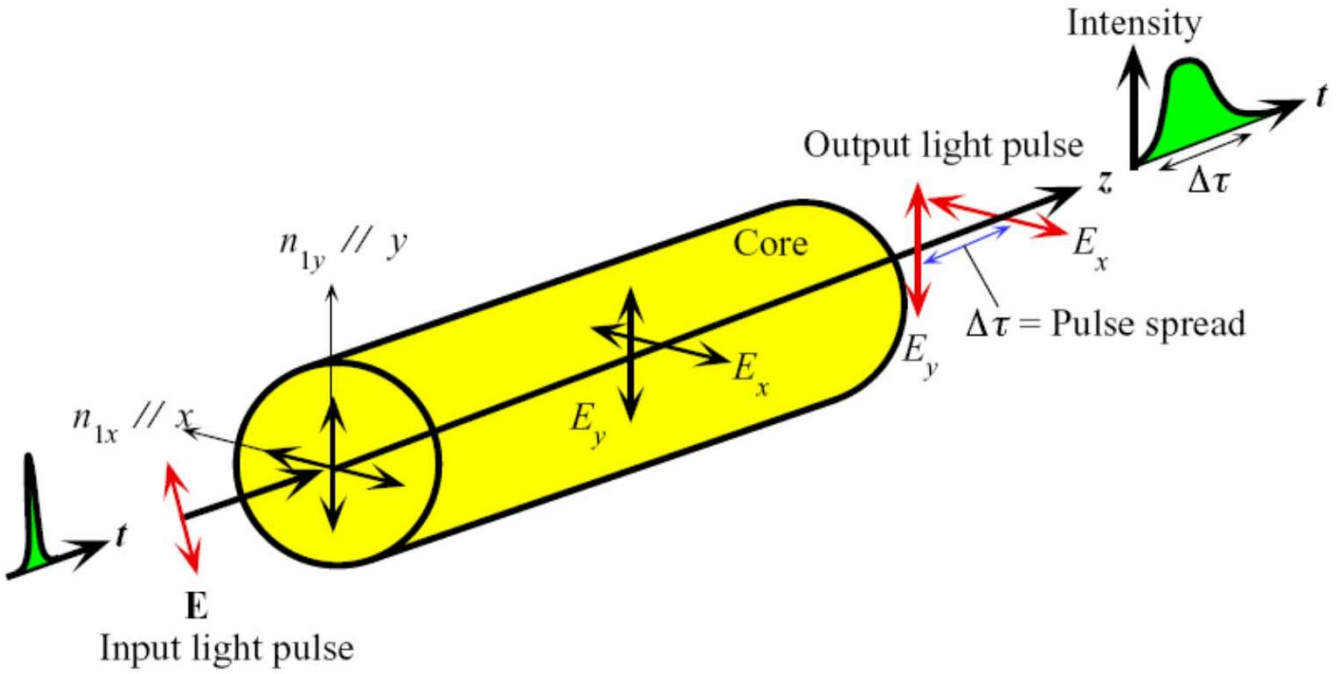
In single-mode fibers, the dispersion of a propagating pulse arises because of the finite width $\Delta\lambda$ of the source spectrum. This type of dispersion caused by a range of source wavelengths is generally termed **chromatic dispersion**, including the material, waveguide, and profile dispersion, since they are all dependent on $\Delta\lambda$.

$$\frac{\Delta\tau}{L} = \left| D_{\mathrm{m}} + D_{\mathrm{w}} + D_{\mathrm{p}} \right| \Delta\lambda$$

$$D_{\mathrm{ch}} = D_{\mathrm{m}} + D_{\mathrm{w}} + D_{\mathrm{p}}$$

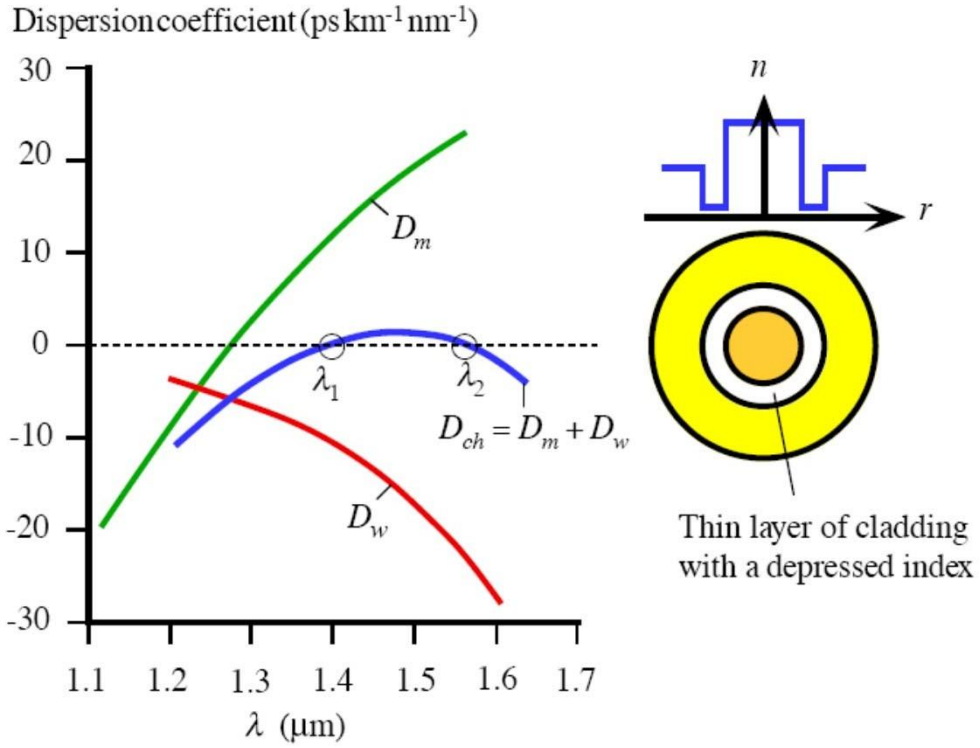
Polarization Dispersion

Polarization dispersion arises when the refractive index is not *isotropic*. When the refractive index depends on the direction of the electric field, the propagation constant of a given mode depends on the polarization. The anisotropic n_1 and n_2 values can result from the fabrication process (changes in the glass composition, geometry, and local strain). Typically, polarization dispersion is less than a fraction of 1 ps km^{-1}. Polarization dispersion **scales roughly with L^2**.



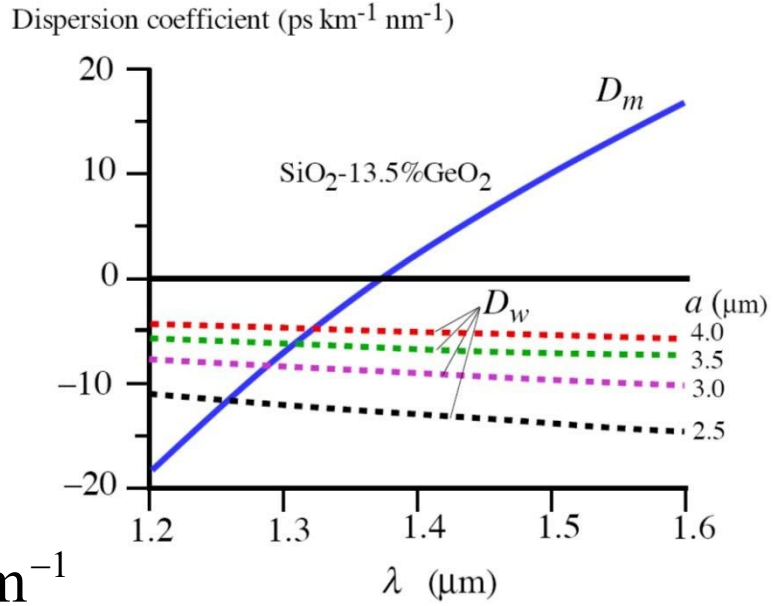
Dispersion Flattened Fibers

D_w can be adjusted by changing the waveguide geometry, for example, using fibers with multiple cladding layers (such fibers are more difficult to manufacture). It is often desirable to have minimum dispersion over a range of wavelengths. For example, fibers with dispersion of 1–3 ps km^{-1} nm^{-1} over the wavelength range of 1.3–1.6 μm allow for wavelength multiplexing, e.g., using a number of wavelengths as communication channels.



Example: Material, Waveguide, and Chromatic Dispersion

A single-mode fiber has a core of SiO_2-13.5 mol%GeO_2 for which the material and waveguide dispersion coefficients are shown in the figure. This fiber is excited from a 1.5 μm laser source with a linewidth $\Delta\lambda_{1/2}$ of 2 nm. What is the dispersion per km of the fiber if the core diameter $2a$ is 8 μm? What should be the core diameter for zero chromatic dispersion at $\lambda = 1.5$ μm?



At $\lambda = 1.5$ μm, $D_m = +10$ ps km^{-1} nm^{-1}
When $a = 4$ μm, $D_w = -6$ ps km^{-1} nm^{-1}
The chromatic dispersion coefficient is

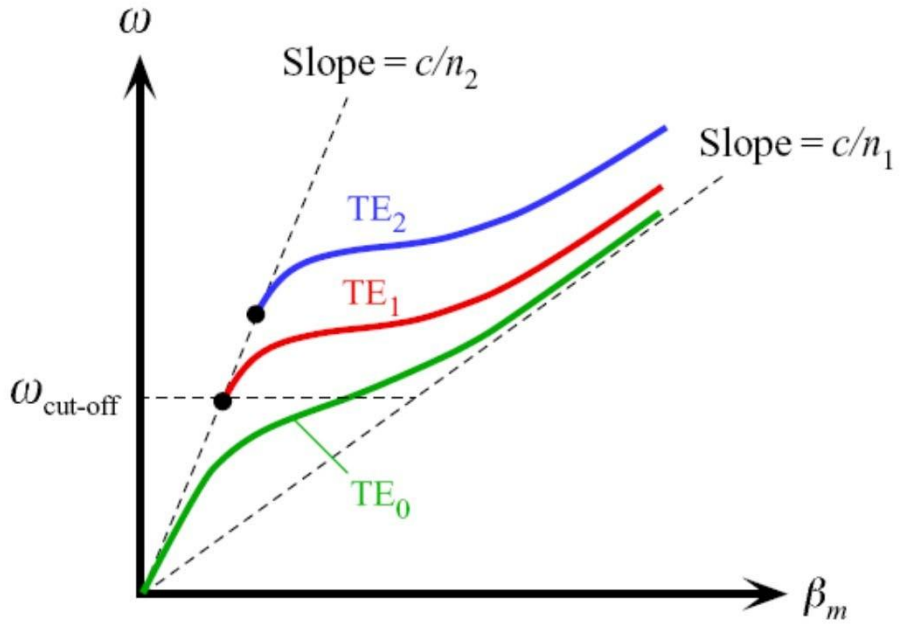
$$D_{ch} = D_m + D_w = 10 - 6 = 4 \text{ps km}^{-1}\text{nm}^{-1}$$

The chromatic dispersion is

$$\Delta\tau_{1/2} / L = |D_{ch}|\Delta\lambda_{1/2}$$
$$= (4 \text{ps km}^{-1}\text{nm}^{-1})(2\text{nm}) = 8 \text{ps km}^{-1}$$

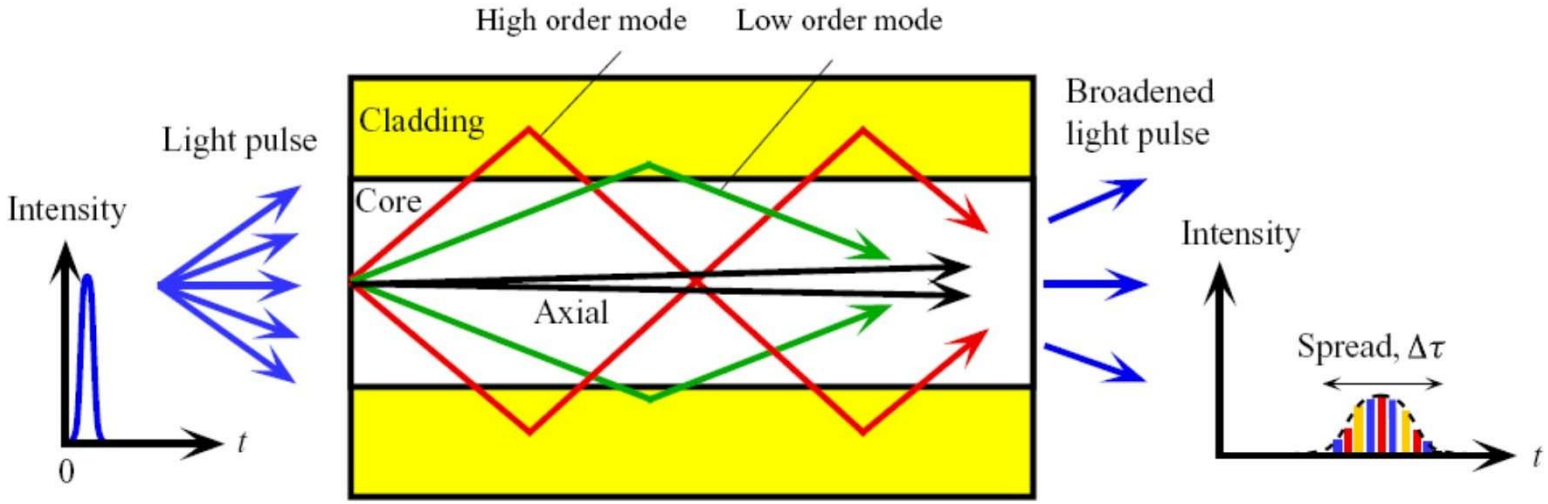
The chromatic dispersion will be zero at 1.5 μm when $D_w = -D_m$ and hence $D_w = -10$ ps km^{-1} nm^{-1}. The core radius should therefore be about 3 μm. **The dispersion is zero only at one wavelength.**

Intermodal Dispersion



The lowest mode (m = 0) has the slowest group velocity, close to c/n_1, because the lowest mode is contained mainly in the core, which has a larger refractive index n_1. The highest mode has the highest group velocity, close to c/n_2 because a portion of the field is carried by the cladding, which has the smaller refractive index n_2.

Different modes take different time to travel the length of the fiber (for an exact monochromatic light wave). This phenomenon is called **intermodal dispersion**.



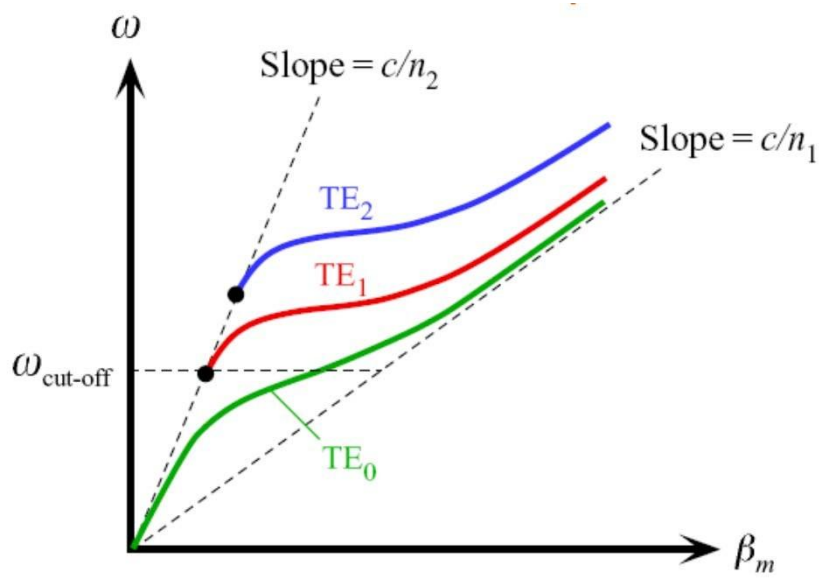
A direct consequence is that a short duration light pulse signal that is coupled into the waveguide will travel along the guide *via* the various allowed modes with different group velocities. The reconstruction of the light pulse at the receiving end from the various modes will result in a broadened signal. The intermodal dispersion can be estimated by

$$\Delta\tau = \frac{L}{v_{g\min}} - \frac{L}{v_{g\max}} \Rightarrow \frac{\Delta\tau}{L} \approx \frac{n_1 - n_2}{c}$$

$$v_{g\min} \approx \frac{c}{n_1},\ v_{g\max} \approx \frac{c}{n_2}$$

$$n_1 = 1.48$$
$$n_2 = 1.46$$
$$\frac{\Delta\tau}{L} = 67\ \text{ns/km}$$

Intramodal Dispersion



The group velocity $d\omega/d\beta_m$ of a single mode changes with the frequency ω. If the light source contains various frequencies (there is no perfect monochromatic wave), different frequencies will travel at different velocities. This is called **waveguide dispersion**.

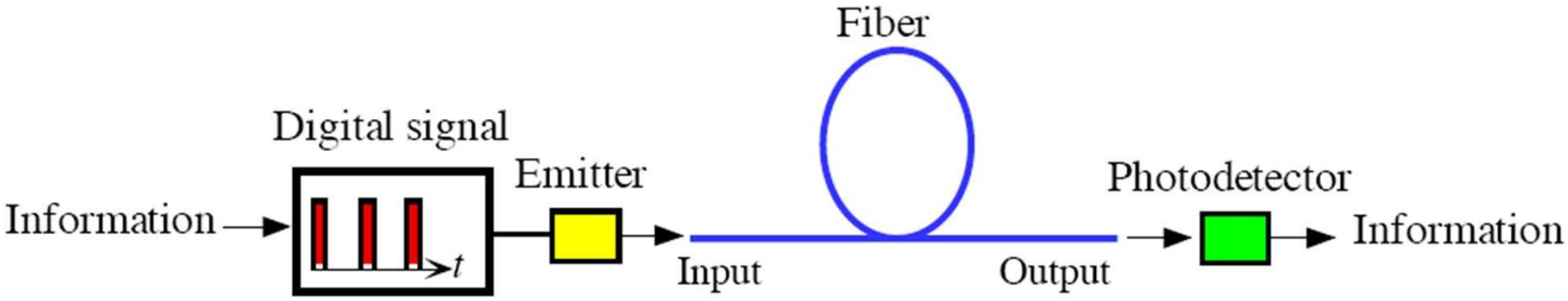
The refractive index of a material is usually a function of the light frequency. The $n(\omega)$ dependence also results in the change in the group velocity of a given mode. This is called **material dispersion**. Waveguide dispersion and material dispersion combined together are called **intramodal dispersion**.

Group Delay

The transit time τ of a light pulse represents a delay between the output and the input. The signal delay time per unit distance, τ/L, is called the **group delay** (τ_g).

$$\tau_g = \frac{\tau}{L} = \frac{1}{v_g} = \frac{d\beta}{d\omega}$$

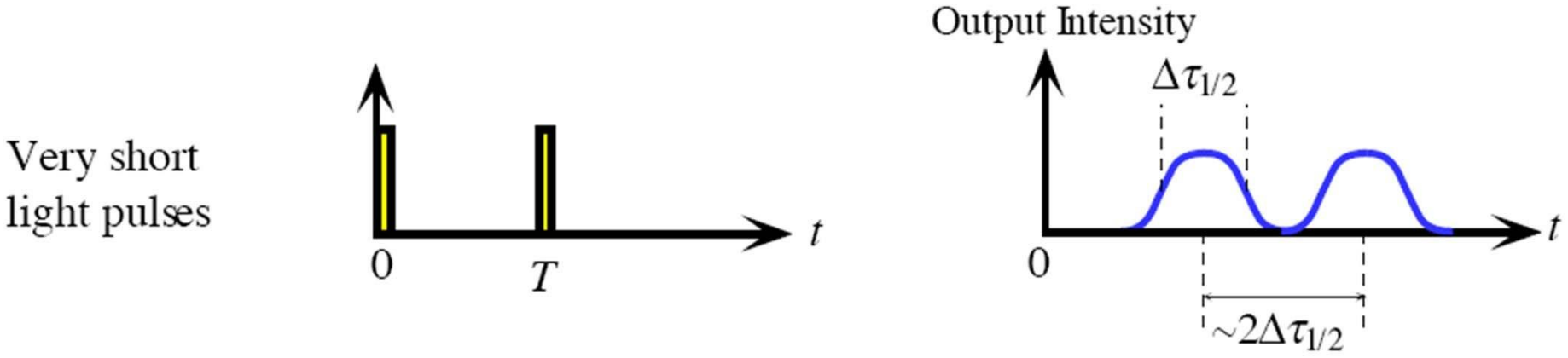
Bit Rate and Dispersion



In digital communications, signals are generally sent as light pulses along an optical fiber. Information is first converted to an electrical signal in the form of pulses that represent bits of information. The electrical signal drives a laser diode whose light output is coupled into a fiber for transmission. The light output at the destination end of the fiber is coupled to a photodetector that converts the light signal back to an electrical signal. The information bits are then decoded from this electrical signal.

Engineers are interested in the maximal rate at which the digital data can be transmitted along a fiber. This rate is called the **bit rate capacity *B*** (bits per second) of the fiber.

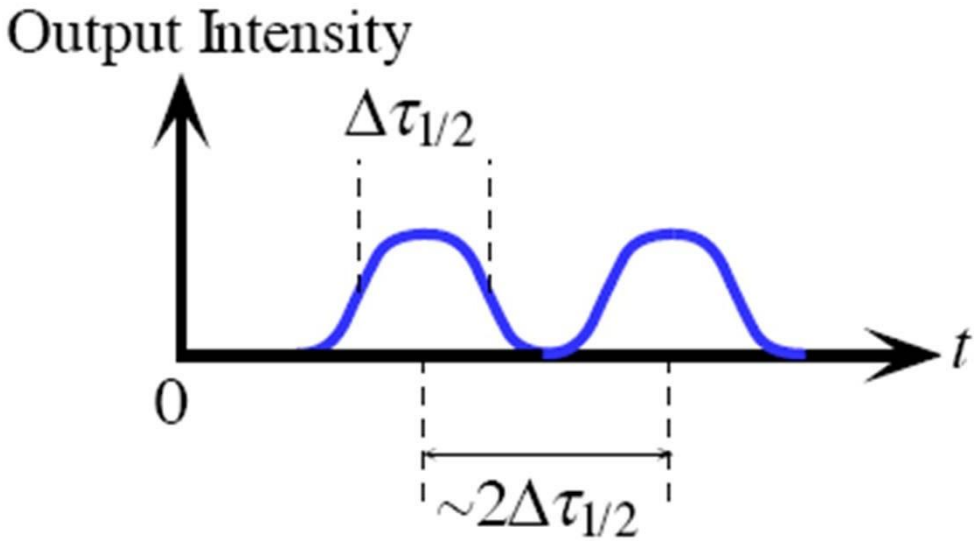
Bit Rate and Dispersion



Suppose we feed light pulses of short duration into the fiber. The output pulses will be broadened due to various dispersion mechanisms. The dispersion is typically measured between half-power (or intensity) points and is called **full width at half power (FWHP)**, or **full width at half maximum (FWHM)**.

To clearly distinguish two consecutive pulses, that is, no **intersymbol interference**, requires that they be separated from peak to peak by at least $2\Delta\tau_{1/2}$ (intuitively).

Bit Rate and Dispersion



There are two types of bit rates. One is called the **return-to-zero (RZ) bit rate**, for which a pulse representing the binary information 1 must return to zero before the next binary information. The other is called the **non-return-to-zero (NRZ) bit rate**, for which two consecutive binary 1 pulses don't have to return to zero in between, that is, the two pulses can be brought closer.

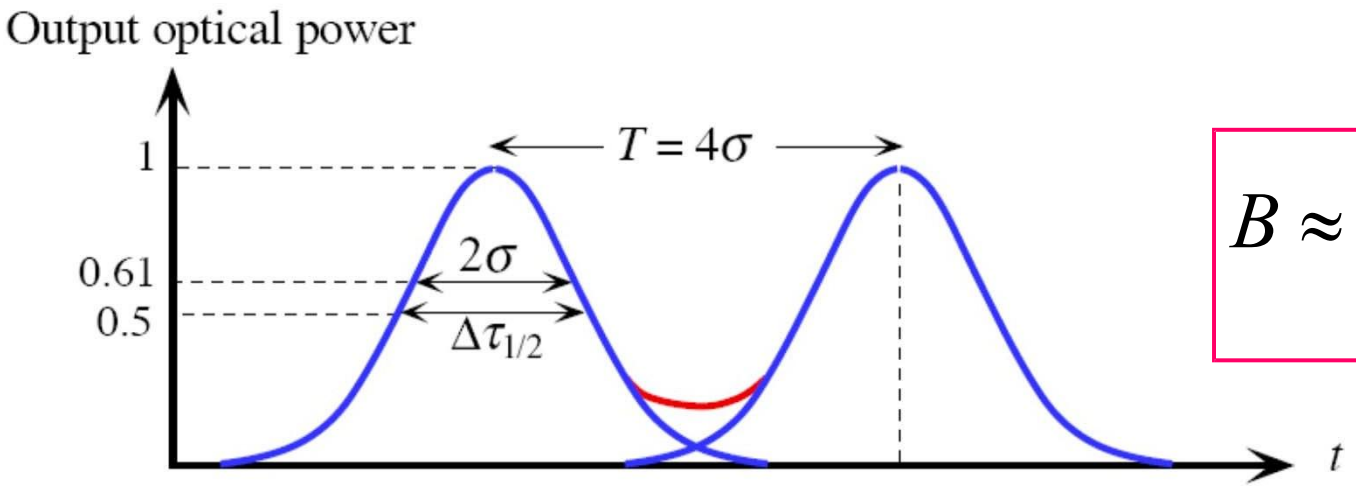
In most cases we refer to the RZ bit rate.

Bit Rate and Dispersion

The maximal bit rate depends on the input pulse shape, fiber dispersion characteristics (hence the output pulse shape), and the modulation scheme of information bits. For Gaussian output light pulses $h(t)$ centered at 0:

$$h(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{t^2}{2\sigma^2}}$$

Standard deviation $\sigma = 0.425\Delta\tau_{1/2}$



$$B \approx \frac{0.25}{\sigma} = \frac{0.59}{\Delta\tau_{1/2}}$$

Bit Rate and Dispersion

Dispersion increases with fiber length L and also with the wavelength range of the source, $\sigma_\lambda = 0.425\Delta\lambda_{1/2}$. It is therefore more appropriate to specify the product of the bit rate B and the fiber length L at the operating wavelength.

$$BL \approx \frac{0.25L}{\sigma_{\text{output}}} = \frac{0.25L}{L|D_{\text{ch}}|\sigma_{\lambda,\text{input}}} = \frac{0.25}{|D_{\text{ch}}|\sigma_{\lambda,\text{input}}}$$

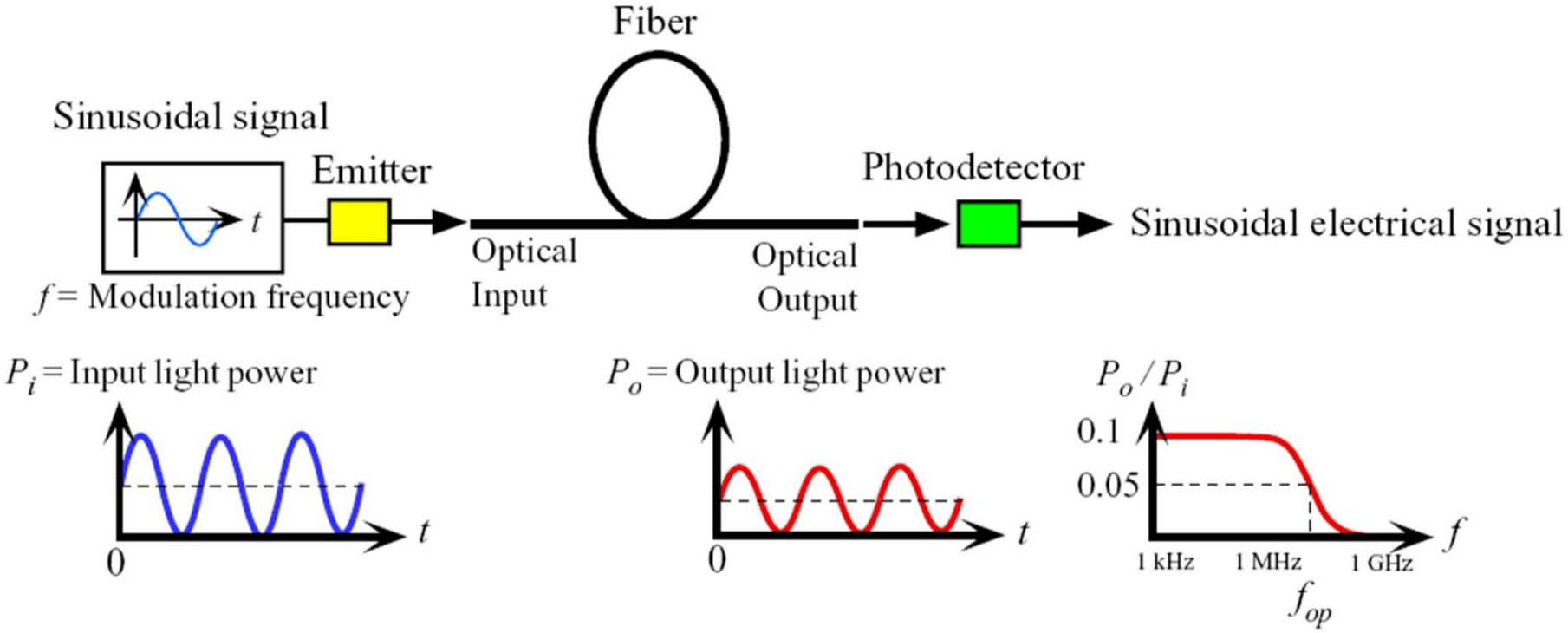
BL is a characteristic of the fiber, through D_{ch}, and also of the wavelength range of the source. In specifications, the fiber length is taken as 1 km and its unit is therefore Gb s^{-1} km.

When both chromatic (intramodal) and intermodal dispersion are present and needed to be taken into account. The overall dispersion can be found according to

$$\sigma_{\text{overall}}^2 = \sigma_{\text{intermodal}}^2 + \sigma_{\text{intramodal}}^2$$

Optical and Electrical Bandwidth

The input light intensity into the fiber can be modulated to be sinusoidal. The light output intensity at the fiber destination should also be sinusoidal with a phase shift due to the time it takes for the signal to travel along the fiber. We can determine the transfer characteristics of the fiber by feeding in light intensity signals with various frequencies.

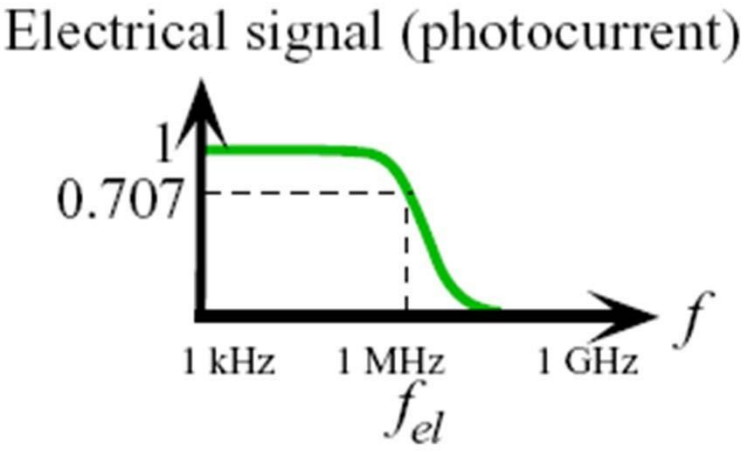


Optical and Electrical Bandwidth

The response, as defined by P_o/P_i, is flat and falls with frequency when the frequency becomes too large so that dispersion effects smear out the light at the output. The frequency f_{op} at which the output intensity is 50% below the flat region defines the **optical bandwidth** of the fiber and hence the useful frequency range in which modulated optical signals can be transferred along the fiber. For Gaussian output light pulses, we have

$$f_{op} \approx 0.75B \approx \frac{0.19}{\sigma}$$

The electrical signal from the photodetector (current or voltage) is proportional to the fiber output light intensity. The **electrical bandwidth**, f_{el}, is usually defined as the frequency at which the electrical signal is 70.7% of its low frequency value.



Relationship between dispersion parameters, maximal bit rates, and bandwidths

Dispersed output pulse shape	FWHM, $\Delta\tau_{1/2}$	B (RZ)	B (NRZ)	f_{op}	f_{el}
Gaussian with standard deviation σ	$\sigma = 0.425\Delta\tau_{1/2}$	$0.25/\sigma$	$0.5/\sigma$	$0.75B = 0.19/\sigma$	$0.71f_{op} = 0.13/\sigma$
Rectangular with full width ΔT	$\sigma = 0.29\Delta T = 0.29\Delta\tau_{1/2}$	$0.25/\sigma$	$0.5/\sigma$	$0.69B = 0.17/\sigma$	$0.73f_{op} = 0.13/\sigma$

Example: Bit Rate and Dispersion

Consider an optical fiber with a chromatic dispersion coefficient 8 ps km^{-1} nm^{-1} at an operating wavelength of 1.5 μm. Calculate the bit rate-distance product (BL), and the optical and electrical bandwidths for a 10 km fiber if a laser diode source with a FWHP linewidth $\Delta\lambda_{1/2}$ of 2 nm is used.

The FWHP dispersion at the output side is

$$\Delta\tau_{1/2} / L = \left|D_{ch}\right|\Delta\lambda_{1/2} = \left(8\text{pskm}^{-1}\text{nm}^{-1}\right)\left(2\text{nm}\right) = 16\text{pskm}^{-1}$$

Assume a Gaussian light pulse shape, the RZ bit rate-distance product is

$$BL = \frac{0.25L}{\sigma} = \frac{0.25L}{0.425\Delta\tau_{1/2}} = \frac{0.25}{0.425\times 16} = 36.8\text{Gbs}^{-1}\text{km}$$

The optical and electrical bandwidths for a 10 km distance are

$$f_{\text{op}} = 0.19/\sigma = 0.19/\left(0.425\Delta\tau_{1/2}\right) = 0.19/\left(0.425\times 16\times 10\right) = 2.8\text{GHz}$$

$$f_{\text{el}} = 0.71\, f_{\text{op}} = 0.71\times\ 2.8\ = 2.0\text{GHz}$$